VICTORY 2027

LIGHT

LECTURE-05 PHYSICS

Shailendra sir

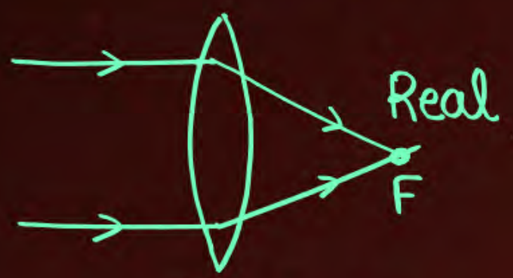


Topics To Be Covered

1. **REVISION**
2. **SIGN CONVENTION**
3. **LENS FORMULA** and Magnification
4. **QUESTIONS**

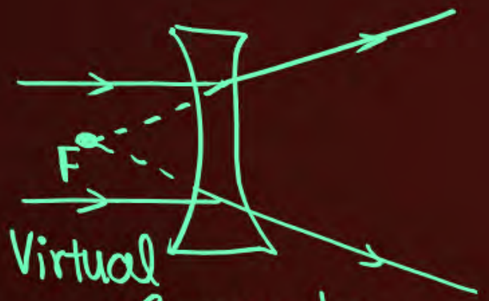

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Physics Wallah

REVISION



Convex Lens

- Real & Inverted
  - Magnified
  - Same
  - Diminished
- Virtual & Erect → Magnified



Concave Lens

Virtual & Erect → Diminished

Real & Inverted → ✗

DIFFERENCE BETWEEN THE IMAGE FORMED BY A CONVEX AND A CONCAVE LENS

HW

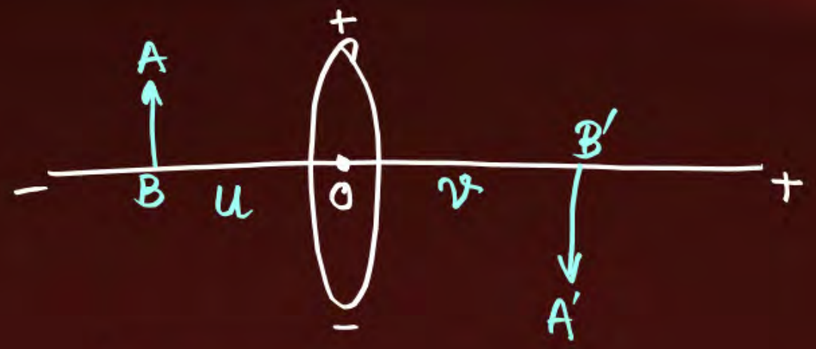
	Image by a convex lens		Image by a concave lens
1.	The image can be real as well as virtual.	1.	The image is always virtual for all positions of object.
2.	The image can be magnified, of same size as well as diminished.	2.	The image is always diminished.
3.	The image can be inverted as well as erect.	3.	The image is always erect.

SIGN CONVENTION

$u \rightarrow$ object distance

$v \rightarrow$ image distance

$f \rightarrow$ Focal Length



$u < 0$

$v > 0$

$f > 0$

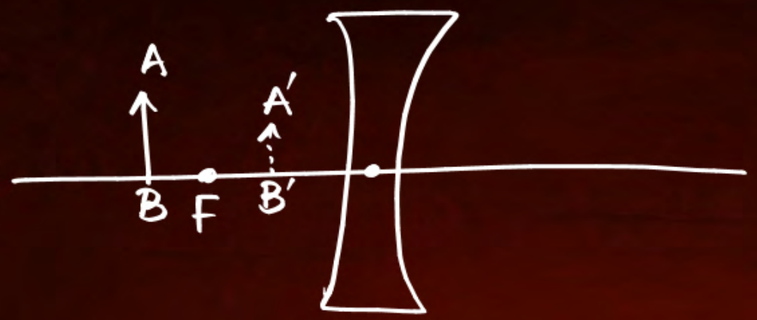
$h_o > 0$

$h_i < 0$



(a) +

(b) −



$u<0$

$v<0$

$f<0$

$h_o>0$

$h_i>0$

LENS FORMULA

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$f_{convex} > 0$

$f_{concave} < 0$

$u < 0$

$v_{real} > 0$

$v_{virtual} < 0$

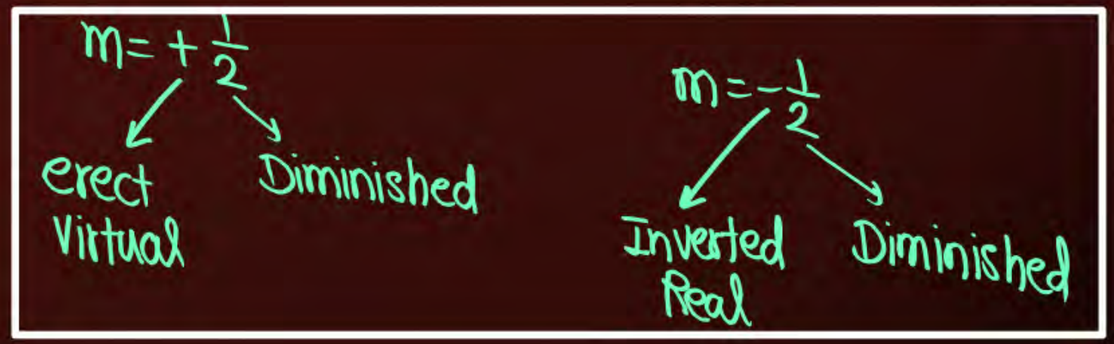
LINEAR MAGNIFICATION

The ratio of length of image I perpendicular to the principal axis, to the length of object O, is called the linear magnification. It is generally denoted by the letter m and is related to the distance of image (v) and distance of object (u) for both the convex and concave lens as :

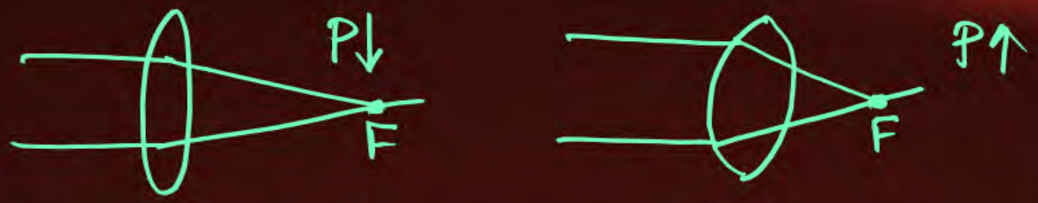
$$m = \frac{h_i}{h_o} = \frac{v}{u}$$







POWER OF A LENS



When a beam of light passes through a lens, it gets deviated from its path. The deviation produced by the lens is expressed in terms of its power. A lens which produces more deviation has more power.

- **The deviation of the incident light rays produced by a lens on refraction through it, is a measure of its power.**
- **A thick lens is of short focal length** and it deviates the rays more, while a **thin lens is of large focal length** and it deviates the rays less.
- Hence power of a lens is expressed in terms of the reciprocal of focal length. Its unit is **dioptre** (symbol D).

$P = \frac{1}{f}$ (in m) or $\frac{100}{f}$ (in cm)

$P_{convex} > 0$

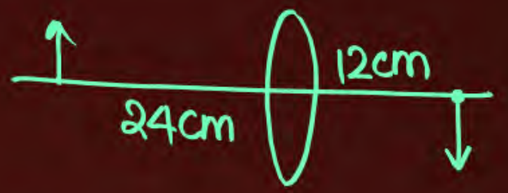
$P_{concave} < 0$

QUESTIONS

An object of height 4.0 cm (h_o) is placed at a distance 24 cm ($u = -24$ cm) in front of a convex lens of focal length 8 cm .

(a) Find the position and size of the image.

(b) State the characteristics of the image.



$h_o = +4$ cm

$u = -24$ cm

$f = +8$ cm

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{8} = \frac{1}{v} - \frac{1}{-24}$

$\frac{1}{v} = \frac{1}{8} - \frac{1}{24}$

$\frac{1}{v} = \frac{3-1}{24}$

$\boxed{v = \frac{24}{2} = 12\text{cm}}$

$m = \frac{h_i}{h_o} = \frac{v}{u}$

$\frac{h_i}{4} = \frac{12}{-24}$

$\boxed{h_i = -2\text{ cm}}$

Diminished

Real

Inverted

QUESTIONS

convex lens

The focal length of a camera lens is 20 cm . Find how far away from the film must the lens be set in order to photograph an object located at a distance 100 cm from the lens.

1. 15 cm
2. 25 cm
3. 35 cm
4. 45 cm

$f = +20\,cm$

$u = -100\,cm$

$v = ?$

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{20} = \frac{1}{v} - \frac{1}{-100}$

$\frac{1}{v} = \frac{1}{20} - \frac{1}{100}$

$\frac{1}{v} = \frac{5-1}{100}$

$v = \frac{100}{4} = 25\,cm.$

QUESTIONS

A convex lens forms an image 16.0 cm (h_i) long of an object 4.0 cm (h_o) long kept at a distance 6 cm from the lens. The object and the image are on the same side of lens.

(a) What is the nature of image?

$v < 0$
Virtual
Erect
Magnified

(b) Find: (i) the position of image, and (ii) the focal length of lens.

$u = -6\,cm$

$$m = \frac{h_i}{h_o} = \frac{v}{u}$$

$$\frac{16}{4} = \frac{v}{-6}$$

$$\boxed{v = -24\,cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{-24} - \frac{1}{-6}$$

$$= \frac{-1+4}{24}$$

$$\frac{1}{f} = \frac{3}{24}$$

$$\boxed{f = 8\,cm}$$

QUESTIONS

If the magnification produced by a lens is −0.5, the correct statement is:

inverted
real

Diminished

1. the lens is concave
2. the image is virtual
3. the image is magnified
4. the image is real and diminshed formed by a convex lens

QUESTIONS

The correct lens formula is

1. $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$
2. $\frac{1}{u}-\frac{1}{v}=\frac{1}{f}$
3. $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$
4. $f=\frac{u+v}{uv}$

QUESTIONS

On reducing the focal length of a lens, its power:

$P = \frac{1}{f}$

1. decreases
2. increases
3. does not change
4. first increases then decreases.

QUESTIONS

$P = \frac{1}{f}$

The lens of power +1.0 D is:

$1 = \frac{1}{f}$

$f = 1\,m$

1. convex of focal length 1.0 cm
2. convex of focal length 1.0 m
3. concave of focal length 1.0 cm
4. concave of focal length 1.0 m.

MAGNIFYING GLASS OR SIMPLE MICROSCOPE

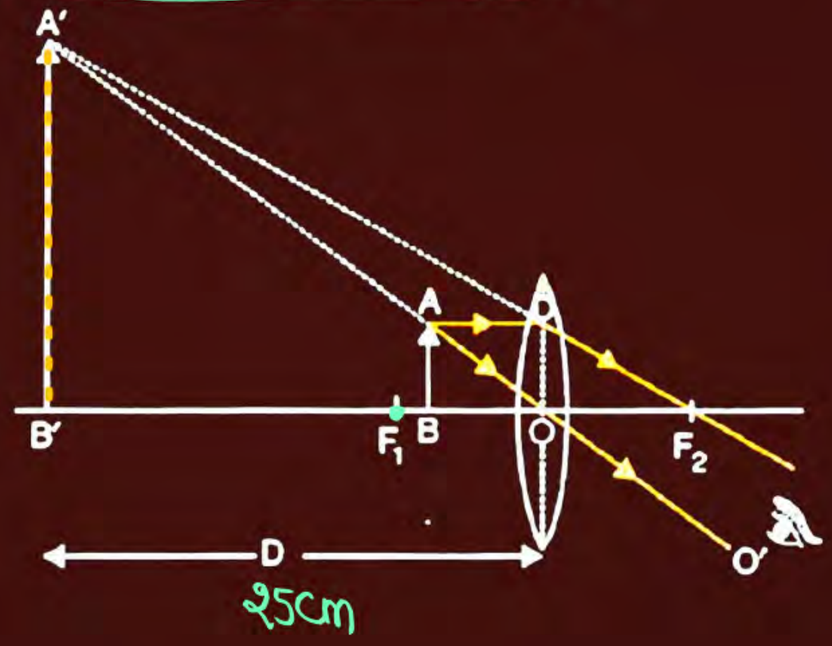
Case I Final image is formed at least distance of distinct vision.

(Virtual, erect & magnified)

$$M = 1 + \frac{D}{f}$$

Case II Final image is formed at ∞.

$$M = \frac{D}{f}$$





QUESTIONS

A magnifying glass forms:

1. a real and diminished image
2. a real and magnified image
3. a virtual and magnified image
4. a virtual and diminished image.

QUESTIONS

The maximum magnifying power of a convex lens of focal length 5 cm can be:

1. 25
2. 10
3. 1
4. 6

$$M = 1 + \frac{D}{f}$$

$$= 1 + \frac{25}{5}$$

$$= 6$$

COLOURS IN WHITE LIGHT WITH THEIR WAVELENGTH AND FREQUENCY RANGE

$v = f\lambda$

Colour	Wavelength Range (nearly)	Frequency Range ($\times 10^{14}$ Hz)
Violet	4000 Å – 4460 Å	7.50 – 6.73
Indigo	4460 Å – 4640 Å	6.73 – 6.47
Blue	4640 Å – 5000 Å	6.47 – 6.01
Green	5000 Å – 5780 Å	6.01 – 5.19
Yellow	5780 Å – 5920 Å	5.19 – 5.07
Orange	5920 Å – 6200 Å	5.07 – 4.84
Red	6200 Å – 8000 Å	4.84 – 3.75

λ_{min} f_{max} λ_{max} f_{min}

NEWTON'S EXPERIMENT

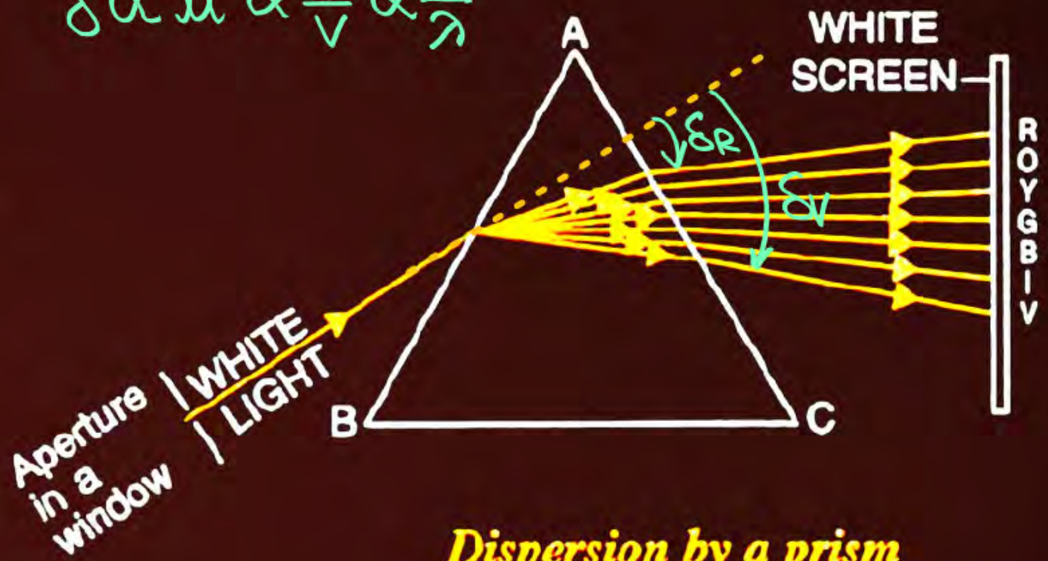
Starting from the side of base of the prism, the order of colours in the spectrum on screen is:

Violet, Indigo, Blue, Green, Yellow, Orange, and Red. This order of colours in the spectrum can easily be remembered by the word **VIBGYOR**.

From the above experiment, Newton concluded that white light consists of seven prominent colours. Each colour corresponds to a small range of wavelength. Thus, white light is a mixture of large number of wavelengths (i.e., it is polychromatic in nature).

$$\delta = (\mu - 1)A$$

$$\delta \propto \mu \propto \frac{1}{v} \propto \frac{1}{\lambda}$$



Dispersion by a prism

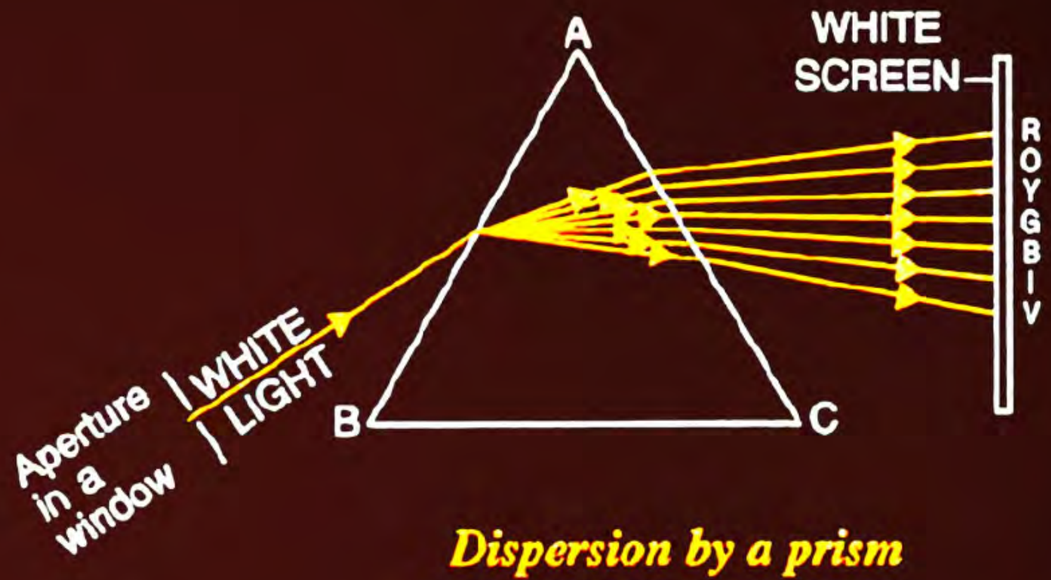
DISPERSION & SPECTRUM

Dispersion

The phenomenon of splitting of white light by a prism into its constituent colours is known as dispersion.

Spectrum

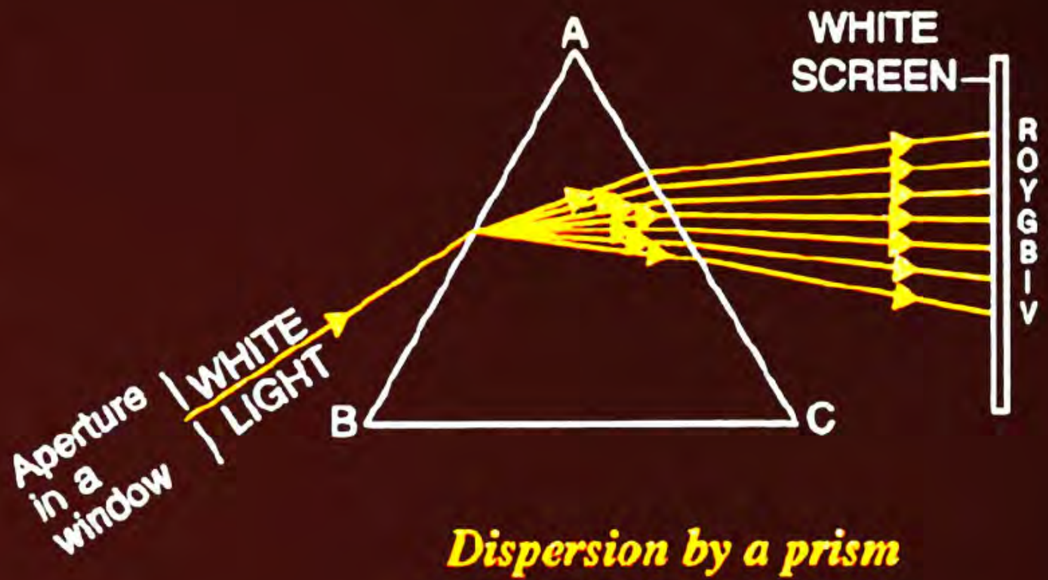
The band of colours seen, on passing white light through a prism is called the spectrum.



Dispersion by a prism

DISPERSION & SPECTRUM

- The cause of dispersion of white light is the change in speed of light with wavelength.
- When white light enters the first surface of a prism, light of different colours due to their different speeds in glass, gets deviated through different angles i.e. the dispersion (or splitting) of white light into its constituent colours takes place at the first surface of prism.
- The violet colour is deviated the most, while the red colour is deviated the least.
- On the second surface of prism, only refraction takes place and different colours are deviated through different angles.
- As a result, the colours get further separated on refraction at the second surface.



Dispersion by a prism

DISPERSION & SPECTRUM

NOTE:

- Dispersion of white light occurs at the first surface of prism. (Refraction)
- Deviation of light occurs at both the surfaces of prism.
- The prism does not produce colours, but it only splits the various colours present in the light incident on it.

QUESTIONS

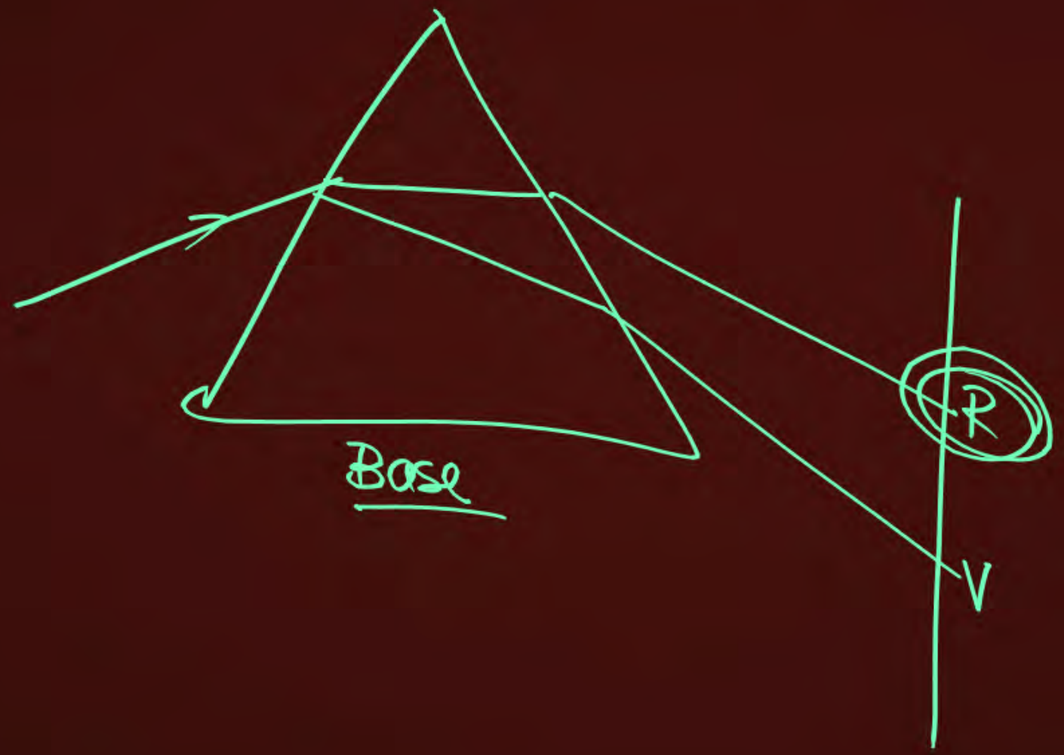
When a white light ray falls on a prism, the ray at its first surface suffers:

1. no refraction
2. only dispersion
3. only deviation
4. both deviation and dispersion.

QUESTIONS

In the spectrum of white light by a prism, the colour at the extreme end opposite to the base of prism is:

1. violet
2. yellow
3. red
4. blue



QUESTIONS

$1A^\circ = 10^{-10}m$ $1nm = 10^{-9}m$

The wavelength range of white light is:

1. 4000 nm to 8000 nm
2. 40 nm to 80 nm
3. 400 nm to 800 nm
4. 4 nm to 8 nm

$4000A^\circ - 8000A^\circ$

$\downarrow$

$400nm \rightarrow 800nm$

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.

Thank You